Automatic Subtitling for Social Media Videos

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Abstract

This document presents an automatic subtitling tool designed for social media videos. The tool leverages OpenAI's Whisper for speech-to-text transcription and FFmpeg for video processing, offering a cost-effective and customizable solution for content creators. It generates subtitles with word-level timestamps, positions captions at the center of the video, and supports advanced styling using ASS files. The tool is user-friendly, supports multiple languages, and provides a free alternative to premium services like Submagic.

1 Introduction

In the era of social media, video content has become a dominant medium for communication. However, many viewers watch videos without sound, making subtitles essential for engagement. To address this, I developed an algorithm that automatically generates and overlays subtitles on videos, tailored specifically for social media platforms. This tool is designed to be cost-effective, customizable, and easy to use, offering a free alternative to premium services like Submagic, which charges $50 per month.

2 Technical Setup

To implement this solution, I utilized a combination of Python-based tools and libraries. The setup process involved the following steps:

2.1 Environment Setup

- **Anaconda**: I used Anaconda to manage Python environments and dependencies, ensuring a seamless integration of the required libraries.
- **FFmpeg**: Instead of installing FFmpeg via package managers (e.g., `choco install ffmpeg`), I opted to install it via `pip`, simplifying the setup process and avoiding additional installations.

2.2 Installation

The installation process involved the following commands:

```
pip install ffmpeg
```

3 Algorithm Overview

The algorithm leverages OpenAI's Whisper for speech-to-text transcription and FFmpeg for video processing. The key features and modifications I implemented include:

3.1 Automatic Subtitling

- The tool automatically generates subtitles for any video using Whisper's transcription capabilities.
- Subtitles are overlaid on the video, making it accessible to viewers who watch without sound.

3.2 Customization for Social Media

- **Captions at the Center**: Instead of traditional subtitles at the bottom, I modified the script to position captions at the center of the video, making them more visually appealing for social media platforms.

- **Word Highlighting**: To enhance engagement, I implemented a feature that changes the color of words as they are spoken, creating a dynamic and interactive viewing experience.
- **ASS Files**: Instead of using SRT files, I utilized ASS (Advanced SubStation Alpha) files for greater flexibility in styling and positioning subtitles.

3.3 Whisper Integration

- I used Whisper's word-level timestamps to break down sentences into smaller groups of words, ensuring precise synchronization between the audio and subtitles.
- The ability to choose different Whisper models (e.g., `small`, `medium`, `large`) allows for improved accuracy, especially for non-English content.

4 Implementation Details

The tool is implemented in Python and consists of several key components. The workflow begins with extracting audio from the video using FFmpeg. The audio is then transcribed using OpenAI's Whisper, which provides word-level timestamps for precise subtitle synchronization. The transcribed text is formatted into ASS files, allowing for advanced styling and positioning of subtitles. Finally, the subtitles are overlaid onto the video using FFmpeg, creating a polished output ready for social media.

5 Usage

The tool is designed to be user-friendly and accessible. Key usage scenarios include:

5.1 Basic Usage

To generate subtitles for a video, the following command is used:

```
auto_subtitle /path/to/video.mp4 -o subtitled/
```

5.2 Advanced Options

- **Model Selection**: Users can specify the Whisper model for transcription. For example:

  ```
  auto_subtitle /path/to/video.mp4 --model medium
  ```

- **Translation**: Subtitles can be translated into English using the `--task translate` option:

  ```
  auto_subtitle /path/to/video.mp4 --task translate
  ```

6 Applications

This tool is particularly useful for content creators, influencers, and marketers who want to enhance their social media videos with professional-looking captions. As a practical example, I plan to use this tool to create engaging content for my stamp collection channel, showcasing my collection while making the videos accessible to a wider audience.

7 Conclusion

By combining OpenAI's Whisper, FFmpeg, and custom Python scripts, I developed a powerful and cost-effective solution for automatic subtitling. This tool not only saves time and money but also provides a high level of customization, making it an ideal choice for social media content creators. With its user-friendly interface and advanced features, it stands as a free and open-source alternative to premium services like Submagic.

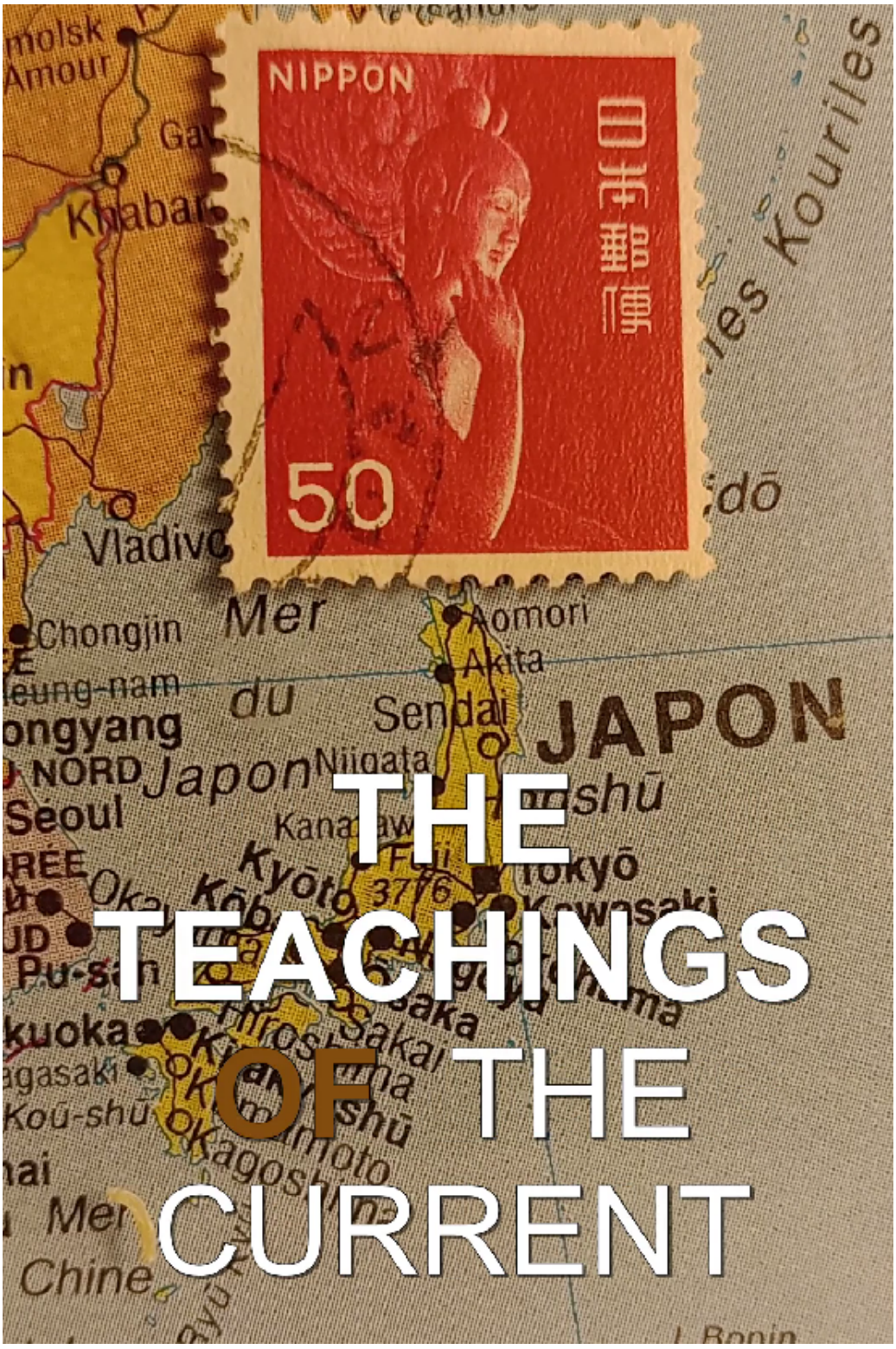


Figure 1: Example of the Automatic Subtitling Tool